// JUSTIN I. CHEN
(858)-436-4774
linkedin.com/in/justin-chnn
chnnjustin@gmail.com

JUSTIN I. CHEN

EDUCATION

University of California, San Diego (UCSD) *Sep. 2023 – Jun. 2027*
Bachelor of Science in Electrical Engineering

- **GPA:** 3.94 / 4.0
- **Relevant Coursework:** C Programming, Object Oriented Programming, Digital Systems and Design, Assembly and Computer Architecture, Circuits and Analog Design, Linear Algebra, Multivariable Calculus, Diff Eq, Vector Calc, Statistics, Physics: (Mechanics, E&M, Fluids, Waves, Thermodynamics, Optics), Signals and Systems, Circuit Components, Probability and Stats, Stochastic Processes, Linear Control Systems, Linear Electronic Circuits, Linear Systems, Linear & Nonlinear Optimization, Digital Comm Theory, Data Networks, Signal Processing, Deep Learning, Machine Learning, Radar Theory and Lab, Digital Communication Lab, Image Processing, Electromagnetism

AWARDS AND HONORS

- **UCSD Jacobs Engineering Scholarship**: Merit based full-ride scholarship, awarded based on academics, leadership, and innovative potential.
- x8 UCSD Provost Honors
- 1st Place Award at 2023 GSDSEF (Greater San Diego Science & Eng. Fair), CSEF Qualifier (CA Science & Eng. Fair)
- AIME Qualifier (American Invitational Math Competition; for high scoring AMC 10/12 participants)
- BMES High School Poster Conference Honorable Mention
- Dean's List Award Semi-finalist at 2021 FIRST Robotics Competition

PUBLICATIONS

- B. Chen, L, Nie, **J. Chen**, W. Zhao, X. Chen, X. Zhang. "*Fingers: Kinematic Aperture mmWave Sensing with a Near-Field Digital Twin*," Submitted to ACM Mobicom 2027.
- B. Chen, **J. Chen**, W. Zhao, K. Zhang, X. Zhang. "*MetaVenom: Field Programmable Passive Metasurfaces*," ACM MobiCom 2026.
- W. Zhao, B. Chen, K. Zheng, X. Chen, W. Zhang, X. Zhang. "*FlowForm: Scalable Passive Metasurface Network for mmWave Coverage Expansion*," ACM SIGCOMM 2026. (**in Acknowledgements**)
- Y. Li, S. Wang, Y. Fu, **J. Chen**, et. al. "*UltraPoser: Pushing the Limits of IMU-based Full-Body Pose Estimation with Ultrasound Sensing on Consumer Wearables*," In ACM Symposium on User Interface Software and Technology (UIST), Sep. 2025

RESEARCH/WORK EXPERIENCE

Yale Department of Electrical & Computer Engineering *Jun. 2026 - Present*
Research Intern under Prof. Tara Boroushaki
mmWaveX: Cross Polarization Radars for Real Time Scene Reconstruction

- Using transformer-based scene reconstruction for 2 cross polarization FMCW radar to derive scene 3d occupancy map.
- Implemented **hardware trigger radar sync on TI AWR1843 and IWR1443** with Arduino sync-in clock.
- Engineered data collection setup for mmWave radar perception with depth camera ground truth and facilitated simultaneous frame firing through strict hardware trigger and ROS supervision.
- Created Flask based dashboard website that facilitated collection debugging, collection, and management.
- Set up and storage server with strict cybersecurity protocols for storing collected data and uploading to compute clusters.
- Implementing **physics-informed sensor fusion with novel positional encoding** similar to RoPE and PRoPE to transformer architecture for flexible NLOS capable scene reconstruction.

UCSD (UC San Diego) Center for Wireless Communications *Oct. 2024 - Present*
Research Intern under Prof. Xinyu Zhang
FinSen: Finger-worn Multistatic RF Sensing using Near-Field Digital Twin

- Researching radar perception for robotics and grasping – enabling flexible material aware perception for NLOS and difficult grasping scenarios.
- Wrote custom bare metal with python and verlog for Infineon BGT60TR13C FMCW radars to facilitate custom SPI trigger commands with FGPA Xilinx Artix 7 FPGA and enable multistatic RSS signal reception for sparse array tomography setup.
- Designed 2d rotational tomography data collection setup with ST2315 servos for FMCW radar angular sweeping.
- Performing object material and geometry joint inference with near-field digital twin solved with differentiable optimization, enabling VLA action models to perform material aware actions.
- Deploying on 2 robot arm and hands and performing real world evaluations, with real-time perception SmolVLA running on NVIDIA jetson, and high accuracy multi-layer differentiation between materials such as sugar, flour, and salt.

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MetaVenom: mmWave Quasi-Passive Metasurfaces for Network Enhancement and JCAS

- Developed new **quasi dynamic passive metasurfaces for mmWaves (60Ghz)** to increase signal coverage that allows for passive reconfigurability using ferrofluid. Our methods reduce are much cheaper than fully active metasurfaces and overcome the weaknesses of passive surfaces not being reconfigurable.
- Implemented and tested joint optimization algorithm using **differential evolution genetic algorithm** that outputs metasurface design and ferrofluid configuration based on incident signal direction and desired output beam patterns. Resulting evaluated beam patterns are close to simulation with around clear **15dB peak RSS increase at designed angles** and multi-angle reconfiguration.
- Presented results at UCSD Summer Research Conference. Published in **Mobicom 2026**.

FlowForm: mmWave Network Coverage Enhancement using Networked Passive Metasurfaces

- Developed passive networked metasurface/reflectarray design and placement optimization alg. for mmWave (60Ghz).
- Tested and validated hierarchical placement and design **optimization algorithm in real world scenarios using gradient descent and differentiable ray tracing on custom FPGA controlled 60Ghz software defined radio** (SDR). Algorithm optimized parameters given environmental constraints, designing for robustness including channel dynamics, interference, multiple users, and nonlinearities, demonstrating a **10+% received signal strength (RSS) + coverage increase from baselines.**
- Presented preliminary results with Poster at DevCom NC4 Technical Review @ UCR. Published in **SIGCOMM 2026**.

UltraPoser: IMU Based Human Pose Estimation Enhanced by Acoustic Sensing

- Researched and implemented multimodal **pose estimation model using IMU and Acoustic Sensing** on consumer wearables: phone, watch, and earbud. Used Acoustic Data to cross validate IMU sensor data, through doppler frequency shift and CIR multipath propagation.
- Designed data collection procedure & preprocessed data for multimodal fusion transformer-based model. Model uses GCN, TCN, Transformers, Cross attention, and multi-modal fusion, achieving **~30% decrease in joint position error** vs IMU only methods.
- Presented Results in 2025 UCSD Undergraduate Research Conference. Published in 2025 ACM UIST.

US Navy: Naval Surface Warfare Center, Philadelphia Division *Jun. 2024 – Sep. 2024*

Research Intern

- Acquired knowledge and expertise in Fluid Dynamics, Reinforcement Learning, and Design Optimization: Implemented a standalone application using MATLAB and Simulink (Signal Processing, Deep Learning, Reinforcement Learning Toolboxes) that used a reinforcement learning and Computational Fluid Dynamics based approach for **axial fan blade design optimization**.
- Employed a customized **Actor-Critic Deep Deterministic Policy Gradient Agent and Finite Difference Method solution of the Navier Stokes Equation** to simulate and optimize fan designs based off parameters such as fluid speed, viscosity, etc.

STEM Extracurriculars

Robotics Club *Oct. 2019 – Jun. 2023*

Vice President and Head of Design for Team 3647

- Constructed an **Odometry robot positioning system using magnetic encoders and IMU sensors**, enabling precise robot navigation & location tracking with centimeter level accuracy, enabling higher task completion rates.
- Developed knowledge in robot control systems, implementing a PID control algorithm for subsystems such as robot arm in order to enable **precise fast non-oscillating sub-degree accuracy motion**. Led to robot winning the Idaho Regional Competition.

Services and Club Activies

- Finance Vice President @ Cornerstone Community Consultants *Oct. 2025 – Present*
- Funding Manager @ Association of Comp. Machinery (ACM), UCSD Student Branch *Nov. 2023 – Jun. 2025*
- Sponsorship Manager @ Eta Kappa Nu (HKN), UCSD Student Branch *July. 2024 – Jun. 2025*
- Membership Chair for Scholar Welcome Society @ UCSD Scholar Society *Oct. 2023 – Jun. 2024*
- CAD and Programming Tutor *Oct. 2022 – Jun. 2023*

Additional Skills

- Computer Aided Design (AutoCAD, OnShape, AutoDesk Fusion 360, Solidworks), CAM (Comp. Aided Manufacturing)
- Fabrication (3D Printing, Milling, Lathing, CNCing)
- Programming (Python, MATLAB, Simulink, Javascript, C, C++, Assembly, MIPS, Bash)
- AI and Machine Learning (OpenCV, Pytorch, SciPy, Numpy, Tensorflow, Pandas, etc)
- Robot Controls with PID, RL, MPC
- Finite Element Analysis, CFD (Fusion 360, Ansys HFSS)
- KiCAD, Eagle, FPGAs, PSpice, LTSpice, Verilog